

CSCD - GROUP 2 - CAPSTONE PROJECT · LOW-LEVEL DESIGN

Ghost Identity Hunter

Automated OSINT Attribution Platform
for Synthetic & Stolen Digital Identities

Version 2.0 · June 2026

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Multi-Vector OSINT Collection

>>

Graph-Based Identity Correlation

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NIST SP 800-53 Aligned Controls

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STIX 2.1 CTI Export

>>

Chain-of-Custody Reporting

Agenda

- 01** Summary & Business Impact
- 02** System Architecture & Topology
- 03** Data Flow & Trust Boundaries
- 04** Threat Modeling — STRIDE & MITRE ATT&CK
- 05** Security Controls — NIST SP 800-53
- 06** Implementation Blueprint & Milestones
- 07** Production-Ready Quality Standards

Team Members

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Business Impact & Objectives

THE PROBLEM

Threat actors construct synthetic identities by compartmentalizing their footprint:

- Burner / VoIP phone numbers (no personal ID required)
- Disposable or forwarding email addresses
- Usernames registered across dozens of platforms
- Stolen or stock profile photographs

Manual OSINT correlation across multiple tools is:

- Time-consuming — hours per investigation
- Error-prone — no quantified confidence
- Inconsistent across analysts
- Missing chain-of-custody audit trail

THE SOLUTION

Ghost Identity Hunter automates the full attribution pipeline:

- ✓ Multi-vector OSINT collection (phone, email, username, image) via BFS orchestration
- ✓ Graph-theoretic correlation links fragmented artifacts into unified persona clusters
- ✓ Quantified confidence scoring per relationship with staleness decay model
- ✓ Legally defensible reports: HTML, PDF, STIX 2.1
- ✓ Zero infrastructure cost — local CLI, SQLite, no cloud dependency

4-Layer Pipeline Architecture

INGESTION LAYER

CLI (Click) → Seed Validation → BFS Orchestrator



COLLECTION LAYER

Phone · Email · Username · Image · Breach Modules



CORRELATION LAYER

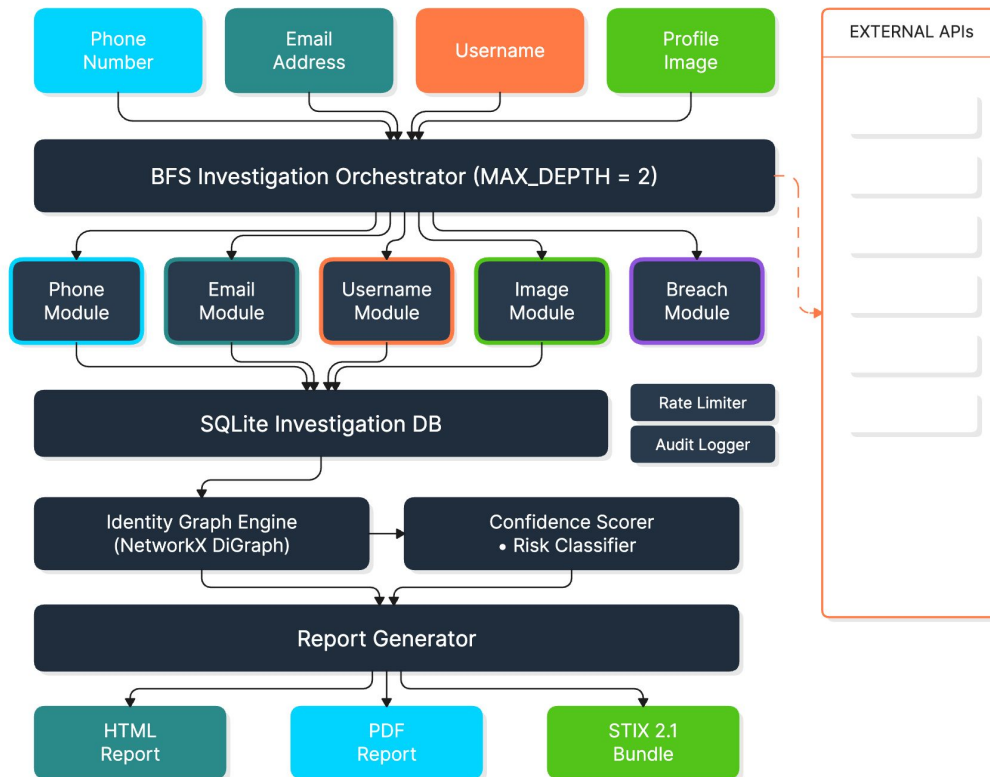
NetworkX Identity Graph → Connected Components → Confidence Scorer



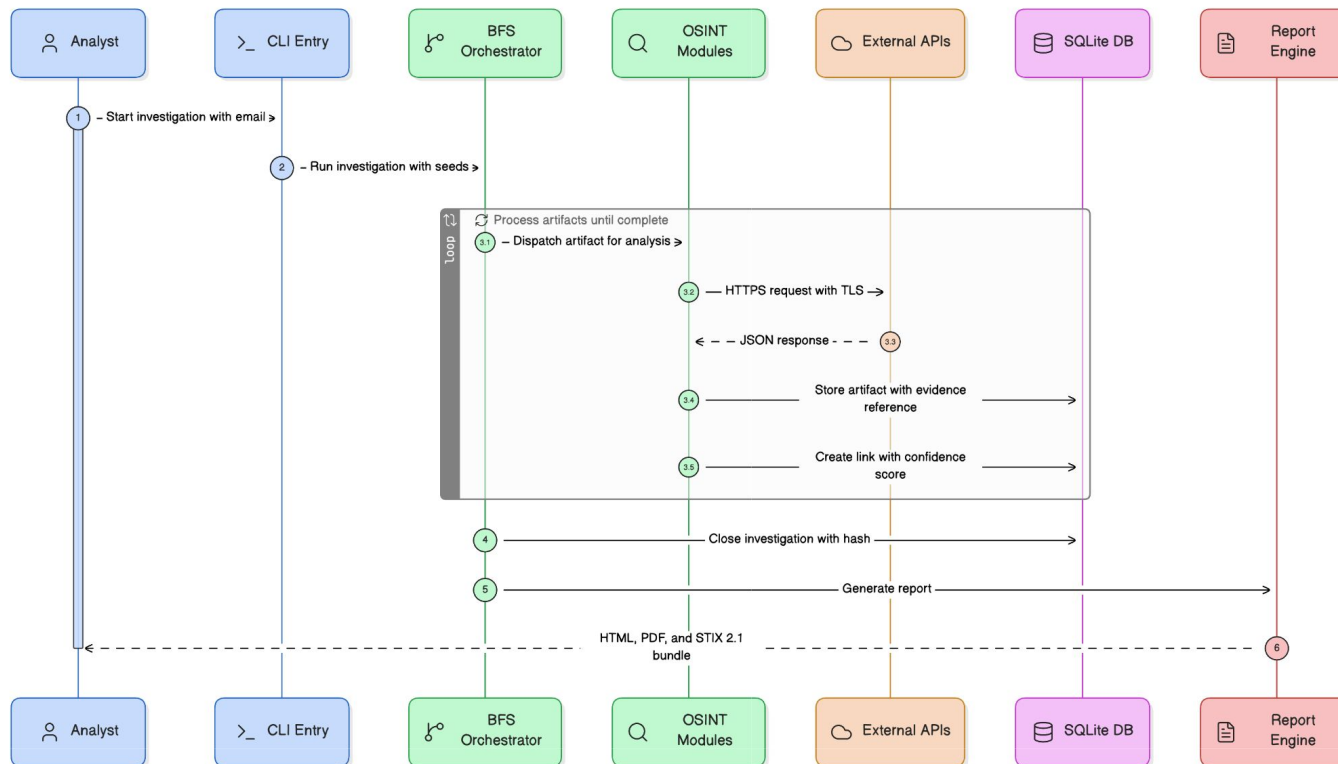
REPORTING LAYER

SQLite Persistence → HTML / PDF Report → STIX 2.1 CTI Export

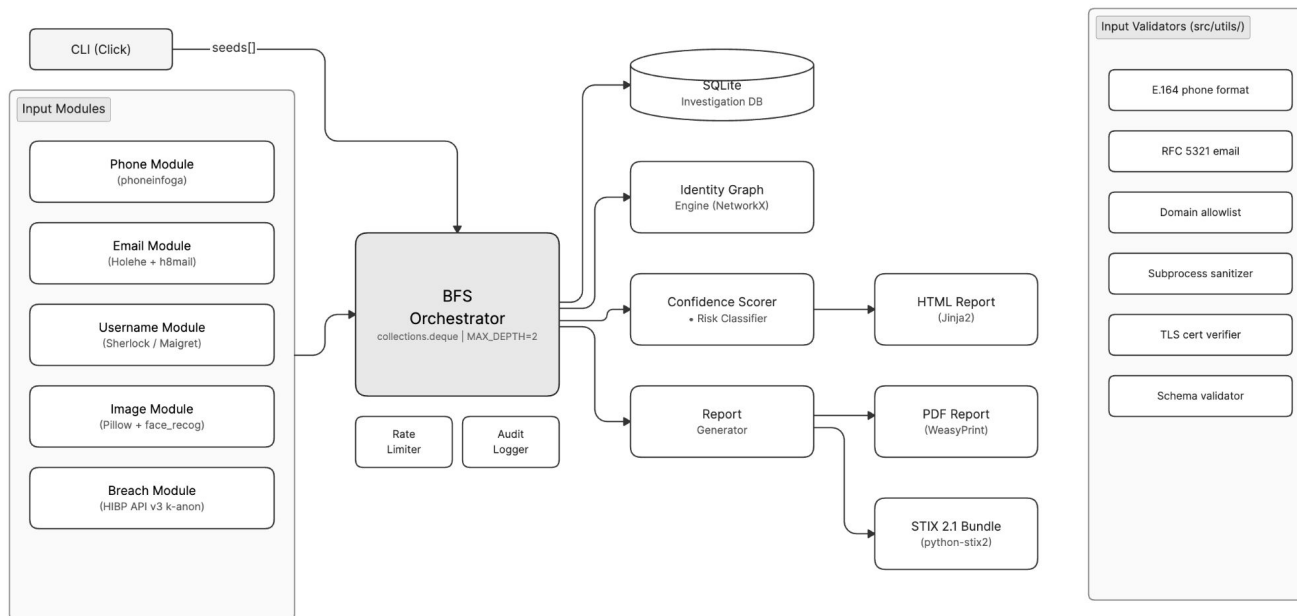
Solution Architecture



Sequence Diagram



Component Interaction



Key Components & Technologies

BFS Orchestrator

Python deque · MAX_DEPTH=2 · per-investigation SQLite isolation

Email Intelligence

Holehe (100+ services) · h8mail · HIBP API v3 (k-anonymity)

Image Intelligence

Pillow (EXIF) · face_recognition (dlib) · Perceptual hash · Reverse search

Confidence Scorer

Typed link weights · staleness decay · cross-type density metric

Phone Intelligence

phoneinfoga (subprocess) · phonenumbers · E.164 · VoIP detection

Username Enumeration

Sherlock / Maigret · httpx · 400+ platforms · false-positive suppression

Identity Graph

NetworkX DiGraph · Connected components · pyvis interactive visualization

Report & Export

Jinja2 HTML · WeasyPrint PDF · python-stix2 STIX 2.1 bundle

400+

Platforms probed

5

OSINT modules

100+

Email services (Holehe)

STIX 2.1

CTI output format

Zero Trust Data Flow

TRUSTED — LOCAL HOST

CLI Input Validation

BFS Orchestrator

SQLite DB (local)

Module Dispatcher

Audit Logger

HTTPS
TLS 1.3
->

UNTRUSTED — EXTERNAL APIs

HavelBeenPwned API

Holehe HTTP Probes

Sherlock / Maigret Probes

WHOIS Servers

Reverse Image APIs

Parsed
Output
->

SANITIZED — PARSED OUTPUT

Schema Validation

No eval() / exec()

Type-checked insertion

Provenance metadata

Evidence chain store

Validated Technology Matrix

Layer	Technology	Version	Justification
Language	Python	3.10+	OSINT ecosystem; async support
CLI	Click	≥8.1.0	Type-safe argument parsing
Phone Intel	phonenumbers / phoneinfoga	≥8.13	E.164 parsing; carrier resolution
Email Intel	holehe + h8mail	Latest	100+ service registration detection
Username Enum	Sherlock / Maigret	Latest	400+ platform parallel probing
Image Analysis	Pillow + face_recog	≥10.0	EXIF extraction; local face similarity
Graph Engine	NetworkX + pyvis	≥3.0	Connected components; interactive viz
Breach Intel	HIBP API v3	—	k-anonymity; no full hash transmitted
Report / Export	Jinja2 + WeasyPrint + python-stix2	≥3.1	HTML / PDF / STIX 2.1
Persistence	SQLite3	stdlib	Zero-dependency; investigator-local
Testing / Lint	pytest + ruff	≥7.0	≥85% coverage; zero lint violations

STRIDE Threat Model — Platform Threats

S	Spoofing Adversary lookalike domain returns false OSINT positives	Medium
T	Tampering Local filesystem access corrupts SQLite evidence chains	High
R	Repudiation No signed audit log; investigator denies query history	Medium
I	Info Disclosure PII in HTTP query params or stdout in CI/CD environments	High
D	Denial of Service Unbounded BFS traversal exhausts external API rate limits	Low
E	Elev. of Privilege Shell metacharacters in OSINT response → subprocess inject	Critical

MITRE ATT&CK Coverage

T1585.001**Establish Accounts — Social Media**

Sherlock/Maigret parallel enumeration across 400+ platforms

T1585.002**Establish Accounts — Email Accounts**

Holehe registration detection; disposable domain classification

T1586**Compromise Accounts**

HIBP breach correlation; stolen credentials to active accounts

T1598**Phishing for Information**

Phone/email artifacts in social engineering from breach data

T1650**Acquire — Prepaid/VoIP Numbers**

phoneinfoga carrier & line-type detection; VoIP flag

T1566.003**Spearphishing via Service**

Profile image reverse search reveals stolen impersonation photos

T1056.003**Input Capture — Web Portal**

Breach data links account credentials to targeted services

NIST SP 800-53 Rev. 5 Control Mapping

PREVENTIVE	SI-10	Input Validation E.164/RFC5321 format checks; subprocess list args only — no shell=True
PREVENTIVE	IA-5	Secrets Mgmt API keys from env vars only; .env excluded from VCS via .gitignore
PREVENTIVE	SC-8	TLS Enforcement httpx verify=True; HTTP-to-HTTPS redirect enforcement; min TLS 1.2
DETECTIVE	AU-2	Event Logging ISO 8601 timestamps; module name; SHA-256 hashed artifact values
DETECTIVE	AU-3	Evidence Provenance source_module + raw_evidence_ref + discovered_at per artifact row
DETECTIVE	SI-7	DB Integrity SHA-256 hash of SQLite DB computed on close; verified on report gen
RESPONSIVE	SC-5	Rate Limit Backoff Exp. backoff with jitter on HTTP 429; max 3 retries per endpoint
RESPONSIVE	IR-4	Evidence Packaging Signed timestamped ZIP: SQLite DB + raw evidence + report on close

Delivery Milestones — 4-Week Roadmap

Phase 1

Wk 1–2 Design & Core Infrastructure

SQLite schema · BFS orchestrator · CLI entry point

Phase 2

Wk 1–2 Collection Modules

Phone · Email · Username · Image · Breach modules + unit tests

Phase 3

Wk 2–3 Correlation Engine

NetworkX graph · Connected components · Confidence scorer

Phase 4

Wk 2–3 Reporting & STIX Export

Jinja2 HTML · WeasyPrint PDF · python-stix2 CTI bundle

Phase 5

Wk 3–4 Integration & Hardening

End-to-end tests · Injection hardening · Rate-limit compliance

SQLite Investigation Schema

investigations

investigation_id TEXT PK
title TEXT
status TEXT CHECK(...)
created_at DATETIME
completed_at DATETIME
db_sha256 TEXT

artifact_links

link_id TEXT PK
source_artifact TEXT FK
target_artifact TEXT FK
link_type TEXT
confidence REAL CHECK(0-1)
evidence TEXT (JSON)

artifacts

artifact_id TEXT PK
investigation_id TEXT FK
artifact_type TEXT CHECK
value TEXT
source_module TEXT
confidence REAL CHECK(0-1)
raw_evidence_ref TEXT

platform_presence

presence_id TEXT PK
platform_name TEXT
profile_url TEXT
username TEXT
is_verified BOOLEAN
raw_evidence_ref TEXT

Identity Graph & Confidence Scoring

IDENTITY GRAPH (NetworkX DiGraph)

Nodes: Artifact records (phone, email, username, image)

Edges: Directed weighted relationships (link_type + confidence score)

Algorithm:

1. Build DiGraph from DB artifacts + links
2. Find weakly connected components
3. Each component = one persona cluster
4. Score = cross-type link density / 5.0 (capped at 1.0)

Risk classification per cluster:

burner_or_voip_phone
disposable_email_address
credential_breach_exposure
stolen_profile_photograph

CONFIDENCE SCORING MODEL

Link Type	Score
exact_match	1.00
registered_with	0.90
breach_linked	0.80
image_match	0.75
username_pattern	0.60
temporal_match	0.40

Staleness Decay: >365 days → ×0.80 | >730 days → ×0.60

Investigation Reports & STIX 2.1 Export

REPORT STRUCTURE

Section	Format	Content
Executive Summary	HTML + PDF	Seed artifacts; persona count; confidence
Identity Profile	HTML + PDF	Consolidated emails, phones, usernames, images
Platform Presence Matrix	HTML + PDF	Profile URLs, creation dates, verification status
Relationship Graph	HTML only	Interactive pyvis force-directed graph
Breach Exposure	HTML + PDF	Breaches per artifact; data types; pwd exposure
Evidence Chain	HTML + PDF	Ordered audit trail with raw evidence references
STIX 2.1 Bundle	JSON file	ThreatActor + Indicators + Relationships + Notes

STIX 2.1 OBJECTS

ThreatActor SDO

One per identified persona cluster

Indicator SDO

Per email, phone, username artifact

Relationship SRO

indicator → indicates → threat-actor

Note SDO

Risk indicators as structured notes

Bundle

All objects in STIX 2.1 envelope

Code Quality, Constraints & Compliance

CODE QUALITY STANDARDS

Type Annotations

mypy --strict enforced in CI

Linting

Zero ruff violations; pre-commit hook

Test Coverage

≥85% line coverage (pytest --cov)

Docstrings

Google style; all public APIs

Dep Pinning

requirements.lock via pip-compile

Secrets Scanning

git-secrets pre-commit; no keys in VCS

OPERATIONAL CONSTRAINTS

BFS Depth

MAX 2 hops

HIBP Rate

≥1,500 ms

Subprocess

30 sec timeout

HTTP Conn

10 sec timeout

DB Warning

500 MB soft limit

Log PII

SHA-256 hashed

LEGAL & ETHICAL

- ✓ Passive OSINT only
- ✓ No credential stuffing
- ✓ No active exploitation
- ✓ GDPR / CCPA / IT Act 2000 compliance
- ✓ API ToS adherence + rate limits
- ✓ Full chain-of-custody audit trail
- ✓ Admissibility-ready evidence packaging

SUMMARY

Ghost Identity Hunter:

- ▶ Automated BFS OSINT collection across 4 artifact types
- ▶ Graph-theoretic persona correlation with quantified confidence
- ▶ STRIDE-modeled security controls mapped to NIST SP 800-53
- ▶ 7 MITRE ATT&CK techniques detected (T1585–T1650)
- ▶ Legally defensible HTML / PDF / STIX 2.1 output
- ▶ Zero infrastructure cost · Passive OSINT only

400+

Platforms enumerated

7

ATT&CK techniques covered

8

NIST controls implemented

\$0

Infrastructure cost